

MAËLYS • HEAD OF AI • HOME ASSIGNMENT

Creative *Loop*

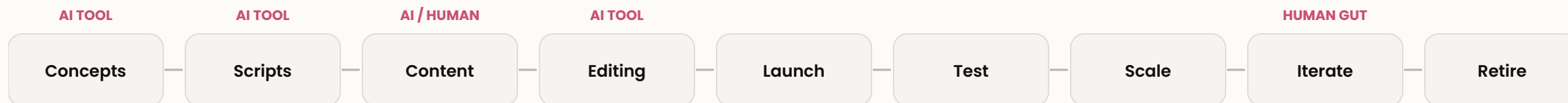
An AI-powered, self-improving acquisition engine — a closed feedback loop between creative production and media performance.

Part 1 • System design

Part 2 • Working prototype: video ad → landing page → next creative

WHERE THE MONEY LEAKS TODAY

Nine steps, each with AI inside — and nothing learning between them



↑ every hand-off is manual · every learning stays in one person's head or one tool's archive

Velocity is bounded by hours

Concept-to-live is gated by human hand-offs, so testing volume — the raw material of learning — is capped regardless of budget.

Learnings evaporate

Why an ad won lives in a media buyer's intuition and a VWO archive. New briefs start from taste, not from the account's accumulated evidence.

Message match decays at the click

From your live funnel: the Caroline LP is byte-identical to the generic advertorial except the H1 and hero image — her ad's promise disappears one scroll in.

The compounding asset isn't the ad. It's the learning.

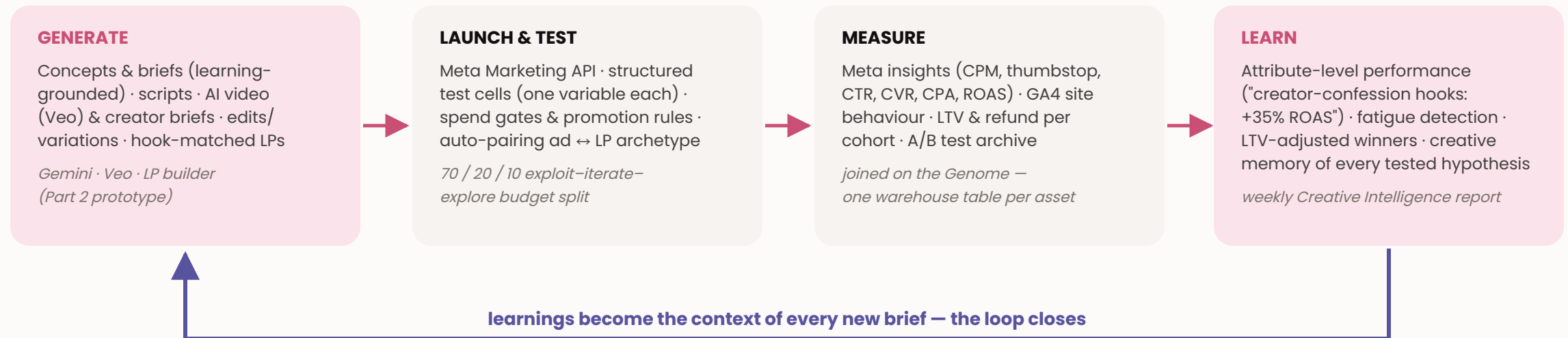
On Meta in 2026, everyone can generate infinite creative. The brands that win are the ones that **learn fastest per dollar of spend** — because targeting is algorithmic and creative is the last controllable lever. So the engine's job is not "make more ads." It is: **make every dollar spent teach the system what to make next.**



Five layers, one loop

CREATIVE INTELLIGENCE — the Creative Genome

Every asset (ad, LP, quiz) is auto-tagged at birth with a structured taxonomy: hook type · angle · persona · format · creator · claims · offer framing · LP archetype. Multimodal models watch every video; tagging is applied retroactively to the entire historical library.



GOVERNANCE — humans stay where humans win

Compliance gate auto-checks claim hedging ("the look of") before anything ships · creative-director approval on new angles · media buyers own overrides. AI drafts; people direct.

The Creative Genome: measure attributes, not ads

A single ad's ROAS is an anecdote. The same spend, measured across a shared taxonomy, is evidence — and it makes your **historical library and A/B archive** a training set from day one, before a single new ad is produced.

GENE	EXAMPLE VALUES (FROM YOUR REAL CREATIVE)	WHY IT PAYS
Hook type	creator confession · problem callout ("GLP-1 loose skin") · result tease · question	Hooks drive thumbstop — the cheapest KPI to move
Angle	post-weight-loss skin · works-while-you-sleep · skeptic conversion	Angles fatigue as a group; detect it before ROAS shows it
Format · Creator	UGC talking head · explainer · montage · Caroline B. · AI-generated	Creator × angle fit is your scaling constraint
Offer framing	TBYB "\$0" · "\$5 trial" · "or don't pay"	Framing moves CVR <i>and</i> refund rate — must be tracked to LTV
LP archetype	advertorial · compact offer-first · creator listicle · quiz	Ad→LP pairing is a free CVR lever nobody measures

Retroactive tagging = instant history

Multimodal models watch every historical ad and LP once (~hours of batch compute) and join the tags to spend you already paid for. The engine is born with memory.

Attribution stays honest

Attribute learnings are directional priors validated by structured tests — never post-hoc certainty. The system proposes; the test confirms.

KEY DATA INPUTS & OUTPUTS

What flows in, what flows out

INPUTS

SOURCE	FEEDS
Historical creatives + spend	Retro-tagged Genome; the engine's starting memory
Meta insights (CPM, thumbstop, CTR, CVR, view time, CPA, AOV, ROAS)	Attribute learnings; test/scale/retire rules
GA4 site behaviour	LP scroll/section engagement → page-level iteration
LTV + 1st-party purchase data	LTV-adjusted CPA target; refund-rate guardrail per offer framing
A/B test archive	Validated learnings seeded into creative memory
Ads Library + competitor sites	Angle-gap map: what competitors run that we've never tested

OUTPUTS

ARTIFACT	CONSUMED BY
Learning-grounded briefs & scripts	Creative team & creator network
AI video teasers (Veo) for hook pretests	Media buying — cheap kill-fast testing
Hook-matched landing pages per ad	Web/CRO (Part 2 prototype does this today)
Weekly Creative Intelligence report	Growth team + leadership — human-readable learnings
Launch/retire recommendations	Media buyers (approve/override)
Creative memory (every hypothesis + result)	The next generation cycle — the loop itself

Four concrete feedback mechanisms

1 · LEARNINGS IN THE PROMPT

The brief generator is literally prompted with the current learning table:
"creator-confession hooks +35% ROAS; explainer format -20%; quiz LP wins for problem-aware traffic." Every new concept must state which learning it exploits — or declare itself an exploration. No more starting from taste.

2 · EXPLORE / EXPLOIT BUDGET

70% of creative slots iterate proven winners, 20% test new hooks on proven angles, 10% are deliberate swings (new angle/format). The split is enforced by the concept queue itself — so the account never overfits to today's winner and never starves learning.

3 · FATIGUE TRIGGERS REFRESH

CTR/thumbstop decay per **angle** (not just per ad) triggers refresh briefs automatically while the winner still profits: same proven angle, new hook and creator — launched before the decay reaches ROAS.

4 · OPTIMIZE TO LTV, NOT FIRST PURCHASE

You sit on LTV data most brands don't have. Winners are re-ranked on 90-day LTV:CAC and refund rate per offer framing — critical with TBYP, where a "\$0" frame can win CVR while quietly leaking into returns and chargebacks.

Learn from history first. Generate second. Automate last.

Sequenced by risk: each phase pays for itself before the next adds autonomy.

CRAWL · WEEKS 1–6

Retroactive intelligence

- Creative Genome schema + auto-tagger over the historical library
- Join tags ↔ Meta / GA4 / LTV / A/B archive in the warehouse
- First weekly Creative Intelligence report

Zero workflow change. Pure insight from money already spent — and the fastest way to earn the team's trust.

WALK · WEEKS 6–16

Learning-grounded generation

- Brief & script generator with learnings in context
- Ad→LP builder in production (Part 2 prototype hardened)
- Veo teaser pretests for hook validation
- Human approval gates on everything

Velocity ×3–5 at constant headcount; the team edits instead of drafting.

RUN · WEEK 16+

Closed-loop automation

- Launch/test/retire rules via Meta API (buyer keeps override)
- Explore/exploit allocation formalized (bandit-style)
- LTV-objective optimization; quiz & LP personalization from ad context

The loop runs nightly. Fitting, for a brand whose product works while you sleep.

Ordered by evidence-per-dollar, not by glamour

#	BUILD	WHY THIS, WHY NOW	EFFORT → PAYOFF
1	Creative Genome + retro-tagger	Converts your entire ad history and A/B archive into structured evidence. Everything else consumes this — and it requires no one to change how they work.	2–3 wks → the engine's memory
2	Learning engine + weekly report	Attribute-level learnings, fatigue alerts, angle-gap map vs competitors. Makes the system's intelligence visible and challengeable by the team from week one.	2 wks → trust + direction
3	Brief & script generator	The highest-leverage generation point: concepts grounded in learnings, in MAËLYS voice, compliance-hedged by construction. Creators create; the engine aims.	2 wks → velocity ×3 at step 1–2
4	Ad→LP builder PROTOTYPE BUILT	Message match is the biggest post-click CVR lever — your own funnel shows hook-match decaying one scroll in. This is Part 2, working today.	hardening only → CVR on every ad
5	Iteration & pretest engine	Auto-variations of winners + Veo teasers to kill weak hooks for dollars before paying creators for full production.	3 wks → cheaper, faster testing

Measure the engine, the business, and the failure modes

ENGINE HEALTH

- **Creative hit rate** — % of new creatives beating portfolio-median ROAS at the test-spend gate (e.g. \$500)
- **Concept→live time** — days, per creative, from queue timestamp to first impression
- **Learning velocity** — validated learnings per month (hypothesis + confirming test)
- **Cost per winning creative** — total test spend ÷ winners promoted

BUSINESS

- **Blended CAC & ROAS** — Meta attribution cross-checked against GA4 + first-party purchase joins
- **90-day LTV:CAC by creative cohort** — the number that decides scaling, not day-1 ROAS
- **New-creative share of spend** — dependence on aging winners is a leading risk indicator
- **Post-click: ATC rate & LP CVR per archetype** — from GA4 events per generated page

GUARDRAILS

- **Refund / chargeback rate per offer framing** — TBYB "\$0" framing must not buy CVR with returns
- **Fatigue index** — thumbstop & CTR decay slope per angle
- **Compliance flag rate** — unhedged claims caught pre-launch (target: 100% caught, 0 shipped)
- **Brand consistency score** — sampled human QA of generated copy per week

I built the loop — it runs today, and it shows its sources

A Next.js app in three stages: drop in any Meta video ad (real MAËLYS creative included)

→ live Gemini video analysis → generated landing page → a real AI-generated next ad.

Every data source the assignment lists is consumed — and labeled where it's used.

1 · Ad Intelligence

Gemini watches the full video natively — audio, on-screen text, pacing — and returns the creative DNA: hook + type, angle, promise, persona, awareness stage, claims, **compliance flags**, recommended LP archetype.

2 · Hook-matched landing page

Real archetypes, voice, offer mechanics and **live CDN imagery** from a teardown of your five LPs — with creator-likeness gating (Caroline's photos only on Caroline ads). The growth team can add free-text creative direction. Every section carries a **provenance chip**: ad analysis · maelys.com · internal data · competitor · your direction.

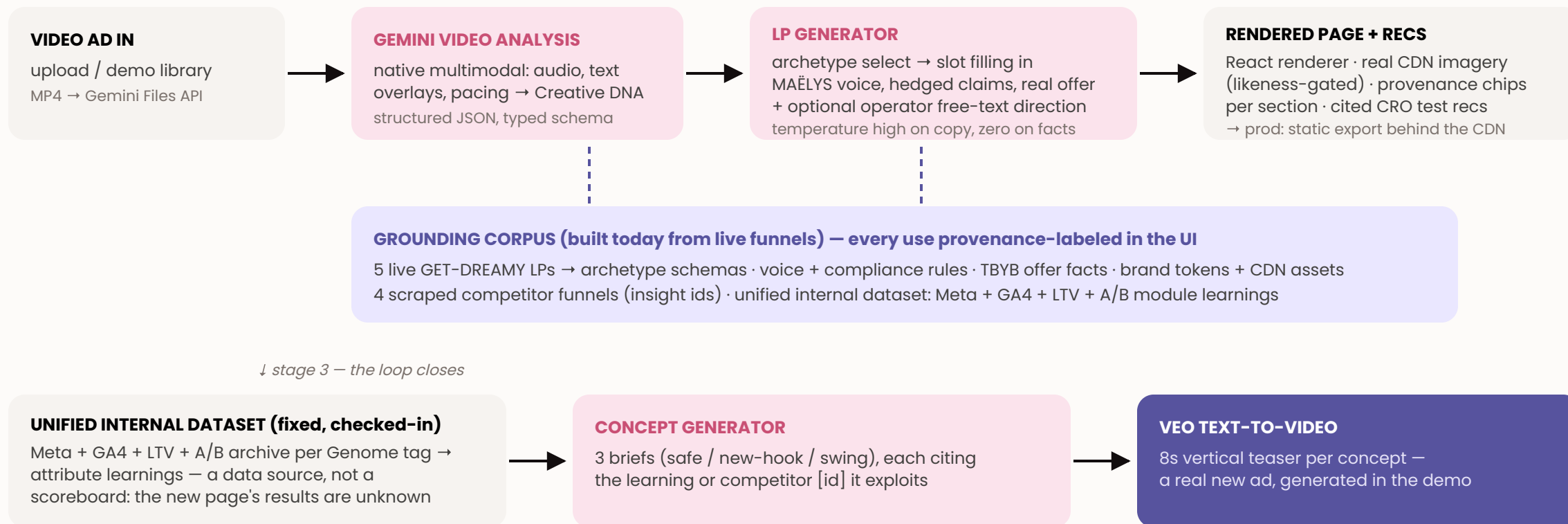
Data as fuel, not scoreboard

The new page's performance is unknown until it runs — so the unified dataset (Meta + GA4 + LTV + A/B archive, fixed & checked-in) and **4 competitor funnels scraped live** (The Body Firm, Truly, Sol de Janeiro, StriVectin) are consumed as cited sources, with clickable links to each competitor page.

3 · The loop closes — with Veo

Attribute learnings + competitor angle gaps prompt three next-iteration concepts (safe / new-hook / swing), each citing what it exploits — and one click generates a real 8-second AI video teaser with Google Veo.

Grounded generation, not blind generation



Six findings from the GET-DREAMY teardown

Hook-match decays after the hero

The Caroline advertorial differs from the generic page by H1 + banner only — zero Caroline below the fold, though your own listicle already has the deep modules to transplant.

Speed leaks CVR before copy matters

VWO loads synchronously with 2s tolerance; 12 autoplay 1080p videos; zero lazy-loading on the flagship advertorial (the listicle has it — the fix exists in-house).

Three prices in one viewport

"\$5 to try" badge, "| \$0" button, "\$43.20 \$54 after" sub-line — dissonance at the exact decision moment, and a refund-rate risk hiding inside a CVR win.

Trust numbers disagree

1M vs 2M jars sold; "30 women/56 days" vs "33/28" on the same product; a hardcoded 15:06 countdown that resets every visit — visible to anyone comparing two LPs from two ads.

Friction at the ATC moment

The offer widget forces an unselected radio choice (the error state is pre-rendered); the two long advertorials have no sticky ATC — only the listicle does.

The quiz forgets its promise

It pledges a "\$35 mystery gift," then the result screen never mentions it — and ignores the user's answers. Echoing zones/concerns + restating the gift is a free lift.

The test queue I'd run first

PRI	TEST	HYPOTHESIS	METRIC
1	Deep hook-match on creator pages — auto-generate the full creator variant (quote modules, "Trusted by" block), which the Part 2 builder already does	Continuing the creator's voice past the hero lifts engaged-scroll and ATC for creator-ad traffic	ATC / CVR
2	Performance pass — async VWO, lazy-load below fold, poster-frame videos	Faster LCP recovers the paid clicks currently lost pre-first-paint	LP CVR, CPC efficiency
3	One price frame — "Try it for \$5 shipping · 3 full weeks" everywhere; default the offer widget to a pre-selected option	Coherence beats the "\$0" trick: equal-or-better CVR with lower refund/chargeback drag on LTV	CVR + 90-day refund rate
4	Sticky ATC on the long advertorials (module exists on the listicle)	19-section pages convert mid-scroll readers they currently strand	ATC
5	Quiz keeps its promises — echo the user's zones/concern in the result + restate the \$35 gift; pass answers into the destination LP	Perceived personalization converts the commitment the yes-ladder already built	quiz→LP CTR, CVR
6	Consistency sweep — one jars-sold number, one clinical footnote, real scarcity or none	Cross-LP shoppers stop catching contradictions that undercut the clinical positioning	trust · brand QA
7	Post-trial "empty-jar" guarantee — 60-day money-back after the TBYB charge (The Body Firm's signature; no MAËLYS LP has it)	Extending risk reversal past the trial charge closes the objection TBYB leaves open	CVR + refund rate

CLOSE

In 90 days: the engine is
learning from every dollar
you've ever spent — and
writing the next brief.

Weeks 1–6 · Genome + retro-tagging + first intelligence report

Weeks 6–12 · learning-grounded briefs + LP builder in production

Week 12+ · Veo pretests + automated test cells

The product works while she sleeps. The engine should too.